Solution Requirements

Date	30 September 2026
Team ID	SWTID-2026-3818
Project Name	Auto_Ticket_Classification_Using_Flow_Diagram
Maximum Marks	4 Marks

Functional Requirements (FR) define what the system should do. Non-Functional Requirements (NFR) define how well it should perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users, agents and admin log in with email and password. Passwords are stored as hashes. (Basic login in first version.)	Medium
2	Authorization levels	Three roles: User (raise and view own tickets), Agent (view assigned tickets, correct category), Admin (view all tickets, upload data, retrain model, see reports).	Medium
3	External interfaces	No paid external service is needed. Optional: email/SMTP for notifications and Graphviz for flow diagrams. Training data is read from CSV files.	Low
4	Transactions / data processing	The system accepts ticket text, preprocesses it, predicts the category and priority, stores the result and routes the ticket, all within one request.	High
5	Reporting	Dashboard with tickets per category, average confidence, low-confidence tickets, status summary. Ticket list can be exported as CSV.	Medium
6	Business rules	• Ticket text must be non-empty and at most 2000 characters. • If confidence is below the threshold (0.60), the ticket goes to manual review. • Each category is mapped to one support team.	High
7	Compliance to laws / regulations	Ticket data may contain personal details, so it is stored only in the local database, and passwords are never stored in plain text. No real personal data is used in the training dataset.	Medium
8	Other	Flow diagram is generated for every ticket and displayed on the result page.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Classification result is shown quickly after submission.	Average response time < 2 seconds
2	Scalability	Model and database can handle a growing number of tickets; the app can later move to PostgreSQL and cloud hosting.	Supports at least 50 concurrent users in testing
3	Security & Data Privacy	Input validation, password hashing, API keys and secrets stored in .env, role-based access.	No plain-text passwords; secrets not committed to Git
4	Reliability & Availability	App handles invalid input and model-loading errors without crashing.	Error rate < 1% during testing

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
5	Usability & Accessibility	Simple, responsive layout; clear labels; readable colours.	A new user can raise a ticket in < 1 minute
6	Other (Accuracy, Maintainability)	Classifier must reach good accuracy; code is modular so the model can be replaced.	Accuracy / weighted F1-score >= 85% on test data